

HARSHITA SINGH

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Education

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

2023 – Present

CGPA: 9.00

Experience

FRONTEND DESIGNER INTERN:

May 2025 – June 2025

Centre for Disaster Mitigation and Management (CDMM)

Under Dr. Surendar M

- Completed an intensive summer internship focused on disaster management technologies and analytical problem solving.
- Collaborated with mentors to understand real-world applications of technology in disaster risk mitigation.
- Received Certificate of Appreciation for outstanding contribution and performance.

SOFTWARE FULL STACK INTERN

2025

Machine Vision and Intelligence Lab, NIT Jamshedpur

- Designed and implemented a high-precision architecture for rich text rendering in modern web editors.
- Developed a reusable rendering framework adaptable to multiple web applications.
- GitHub Repository: github.com/Harshita1803/NITPR

Projects

Smart Agricultural Supply Chain System | Full Stack Web Application

Smart India Hackathon Finalist

- Developed a full-stack logistics platform to manage agricultural product transportation efficiently.
- Designed modules for farmers, transporters, and buyers to streamline product movement.
- Improved supply chain workflow by reducing delays and increasing transparency.

Smart Campus Navigator | Web Application

DevJams (GDSC-VIT)

- Developed an interactive campus navigation platform to improve accessibility.
- Integrated location-based navigation and optimized route guidance.
- Live Demo: www.devjams.gdscvit.com
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Safe Driving Monitoring System | Web + Sensors

Caterpillar Hackathon

- Developed a driver safety monitoring system capable of detecting unsafe driving behaviour in real time.
- Integrated sensor data to improve driving awareness and accident prevention.

Tic-Tac-Toe Game | React.js

Gravitas Finalist

- Built a responsive two-player Tic-Tac-Toe application using React.js.
- Implemented efficient game logic with real-time winner detection and restart functionality.

AI & NLP Automation Suite | Python, n8n, LLM, Neo4j

- Developed an NLP-powered workflow using n8n and Large Language Models to retrieve structured information from datasets using natural language queries.
- Built a domain-specific chatbot using Python, TF-IDF Vectorization, Cosine Similarity, JSON Corpus, and NLTK for intelligent question answering.
- GITHUB LINK- <https://github.com/Harshita1803/CHATBOT-1>
- Designed a Knowledge Graph using Neo4j and Cypher queries to perform graph-based reasoning and relationship discovery.
- Implemented corpus creation, text preprocessing, retrieval-based question answering, and workflow orchestration using modern NLP techniques.

Adaptive Checkpoint-Resume Framework for Stateful Execution in Serverless Computing | Node.js, Express.js, Cloud Computing

- Designed a predictive runtime monitoring framework enabling checkpoint-resume execution for long-running serverless workloads.
- Implemented dynamic state serialization and recursive execution to overcome serverless timeout limitations.
- Developed a prototype using Node.js, Express.js and JSON-based state management with integrated API security validation.
- Focused on improving reliability, minimizing recomputation overhead and enabling pseudo-stateful execution in stateless serverless environments.

Underwater Image Enhancement using Digital Image Processing | Computer Vision

- Developed a computer vision-based image enhancement system to improve underwater image quality using Digital Image Processing (DIP) techniques including noise filtering, adaptive histogram equalization, contrast enhancement, and image sharpening.
- Built an interactive web application for real-time visual comparison and performance evaluation of enhancement algorithms, enabling efficient analysis of image quality improvements.

Lightweight Physics-Informed Neural Network for Welding Defect Classification | Edge AI

- Designed a lightweight Physics-Informed Neural Network (PINN) for real-time welding defect classification using public datasets, integrating physics-guided learning with deep learning for improved prediction accuracy.
- Optimized the model for Edge AI deployment on resource-constrained devices, enabling low-latency inference, efficient computation, and reliable industrial quality inspection.

GITHUB LINK-<https://github.com/Harshita1803/serverless-patterns-lab/tree/main/internal/build>

VERCEL DEPLOYMENT LINK- LIVE DEPLOYMENT LINK

<https://serverless-patterns-lab.vercel.app/>

CODING PROFILES:-

- GitHub : <https://github.com/Harshita1803>
- LeetCode : https://leetcode.com/u/HARSHITASINGH_18/
- CodeChef : https://www.codechef.com/users/team_violet_35
206+ Problems Solved | Bronze Problem Solver Badge

Technical Skills

Languages	Java, JavaScript, Python, HTML, CSS
Frameworks	React.js, Node.js, Express.js, Next.js, Tailwind CSS
Databases	MySQL, JSON
Cloud & Tools	Git, GitHub, n8n, Neo4j, VS Code
Core Concepts	Data Structures and Algorithms, Object-Oriented Programming, Problem Solving, NLP, Machine Learning Basics, Cloud Computing, Knowledge Graphs

Leadership Activities

- Senior Core Member – VIT Anchoring Club

- Senior Core Member – VIT Geospatial Club

Interests

Coding, Algorithms, Artificial Intelligence, Cloud Computing, Problem Solving, Reading

